THE UNIVERSITY OF AUCKLAND

SEMESTER TWO 2020
Campus: City, Offshore Online

MECHANICAL ENGINEERING

Acoustics for Engineers

(Time Allowed: THREE hours)

NOTE: This exam contains 5 questions. Answer **all** parts of **all** questions.
All 5 questions are worth equal marks.
Equations are provided in the APPENDICES BOOKLET.

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Ques.	Mark
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Total	

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1. a) An acoustical engineer records a sound played by a loudspeaker as a digital signal using a sound recorder equipped with a single microphone. Figure 1.1 shows the spectrogram of the audio file played from the loudspeaker.

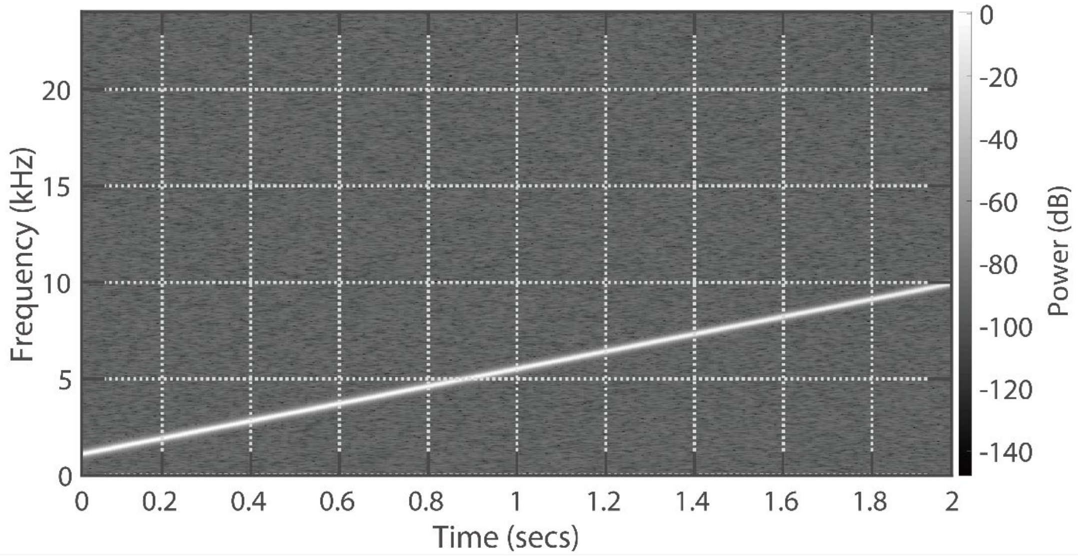

Figure 1.1

i) What is the name of the signal played from the loudspeaker?

(1 mark)

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ii) Assume that the signal is recorded by pulse code modulation (PCM) with 16 bits/sample and the sampling rate of the recording is set at <u>twice the Nyquist rate</u> of the sound. Determine the size of the file saved on the recorder <u>in bytes</u>.

(3 marks)

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Answer to Question 1.a.ii continued.

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iii) When the engineer changes the settings of the recorder and records the same sound, the spectrogram of the recording is changed as shown in Figure 1.2. Explain what changes the engineer has made to the recorder's setting and why it affected the spectrogram of the recording in no more than 100 words.

(3 marks)

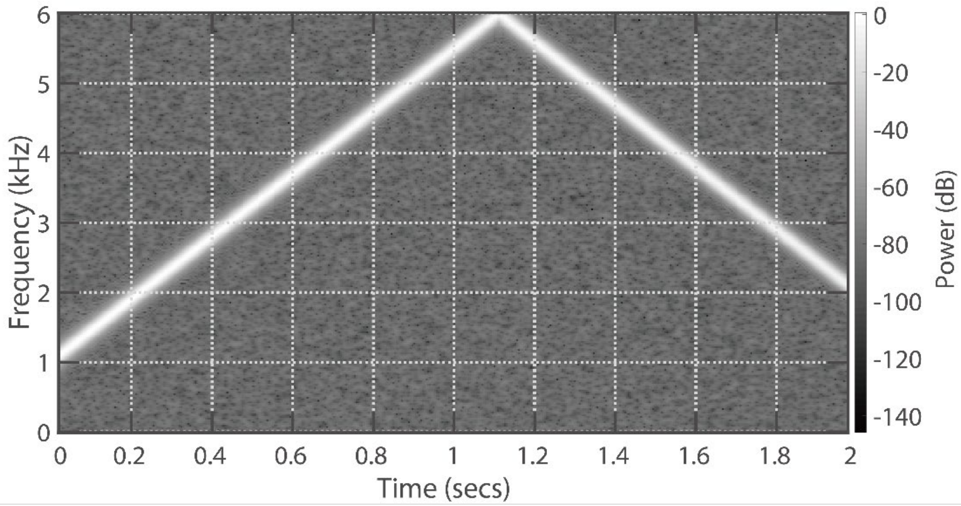

Figure 1.2

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b) Room impulse responses (RIR) were measured at different positions in the room shown in Figure 1.3 using an omni-directional microphone. The distance from the loudspeaker to each position was equal. Positions 2 and 3 were identical, but for Position 3, there was a partition in front of the microphone blocking its line of sight to the loudspeaker. (Note that the partition is not present in Positions 1 and 2.)

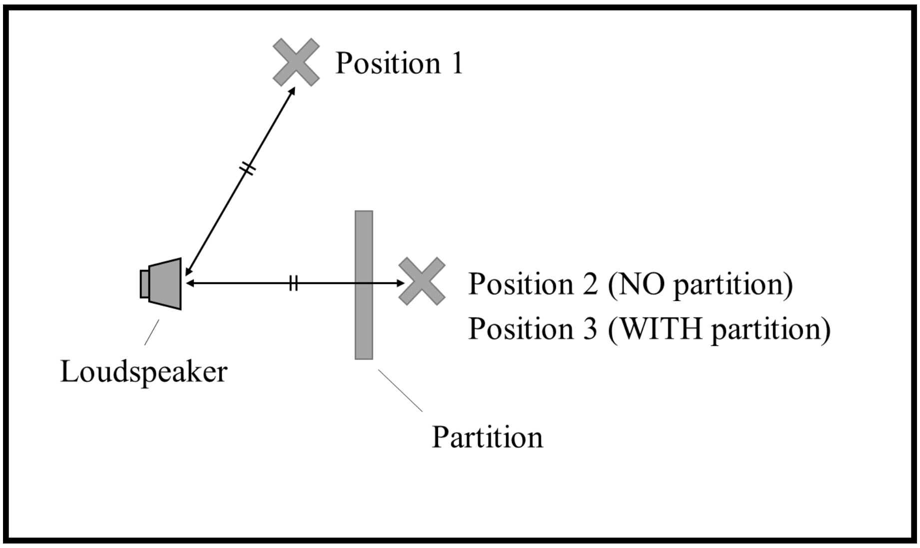

Figure 1.3

i) Determine which of the impulse responses (A) to (C) in Figure 1.4 were measured at the Positions 1 to 3 with a brief justification.

(4 marks)

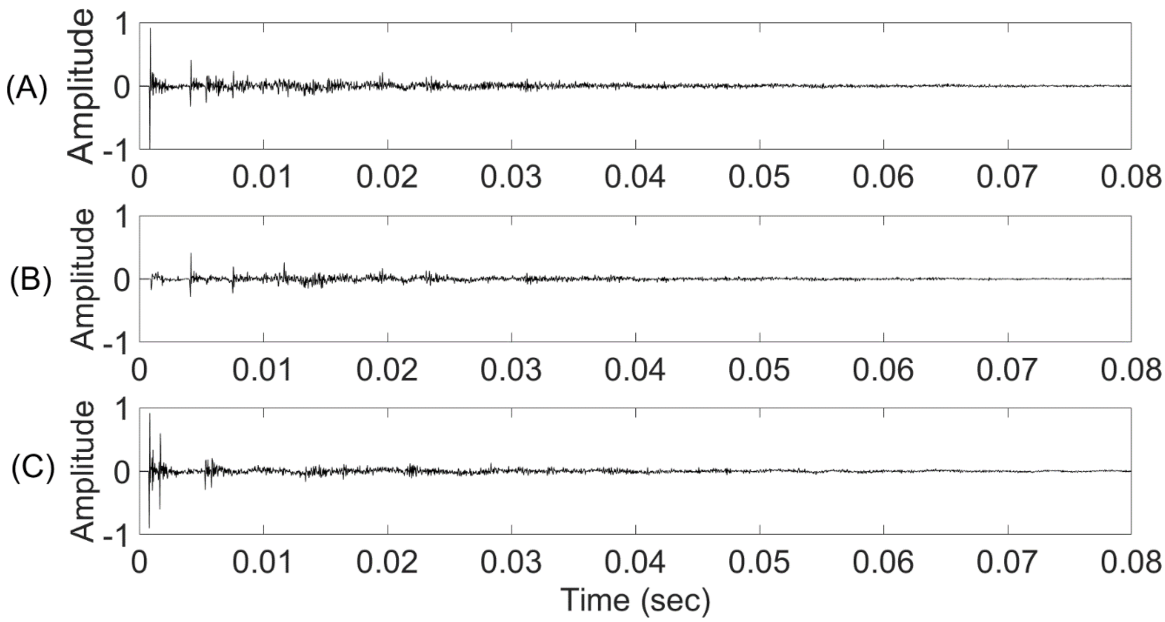

Figure 1.4

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(A)	(B)	(C)

Justification:

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ii) The reverberation time of the room at 1 kHz was calculated from one of the measured RIR using the T30 measure. The following description explains the procedures taken to derive the reverberation time. Fill the blanks (A) to (E).

(5 marks)

"Firstly, a ____(A)____ *filter was applied to the measured RIR. This filter 'passes' frequency components in the subband centred at 1 kHz. If the filtered RIR is denoted as h(t), where t is the time in seconds, the early decay curve (EDC) was calculated by implementing the formula* ____(B)____*. The slope of the EDC was estimated using linear regression, which was applied to the time period between two points* ____(C)____ *dB and* (D)____ *dB on the EDC below the initial level of the EDC. The formula given by the linear regression was y = At + B, where A = -200 dB/s, B = 5 dB and y is the linear fit to the EDC in dB. Hence, the reverberation time of the room at 1 kHz was determined as* ____(E) *seconds."*

(A)
(B)
(C)
(D)
(E)

iii) The measured RIR shown in Figure 1.4 also includes the characteristics of the loudspeaker and microphone used for the measurement. Summarise how one could remove the characteristics of the loudspeaker and microphone from the measurement. Assume the loudspeaker, microphone and room are all linear time-invariant systems.

(4 marks)

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Answer to Question 1.b.iii continued.

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2. a) Figure 2.1 shows a schematic diagram of human ear. Answer the following questions.

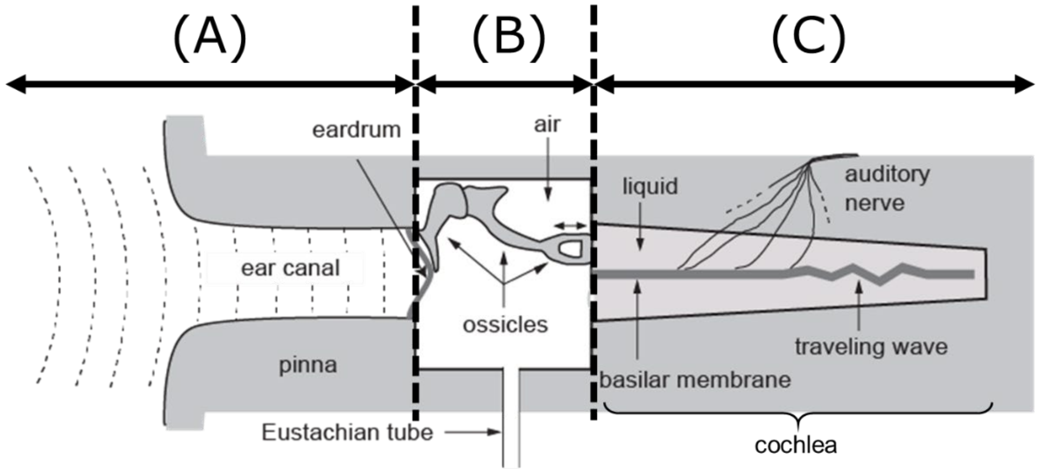


Figure 2.1

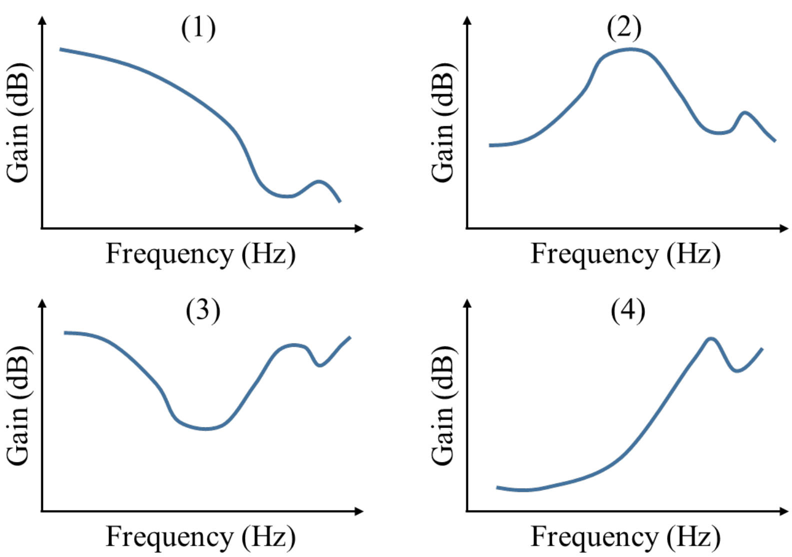
i) Write the name of the regions of the ear labelled (A), (B) and (C) in Figure 2.1.

(3 marks)

(A)
(B)
(C)

ii) When the regions (A) and (B) (i.e. from the entrance of the ear canal to the boundary between (B) and (C)) are modelled as a linear time-invariant system, which of the following diagrams (1) to (4) best describes the shape of its frequency (amplitude) response? Circle the correct diagram.

(2 marks)



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iii) Explain how the cochlea transforms the mechanical vibration of sound to neural signals by referring to the roles of the following parts of the cochlea: basilar membrane, Organ of Corti, inner hair cells, outer hair cells, <u>in no more than 200 words</u>.

(5 marks)

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Answer to Question 2.a.iii continued.

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b) List <u>ALL factors</u> that affect speech intelligibility. Identify the factor(s) that can be quantified by the speech transmission index (STI).

(3 marks)

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c) Figure 2.2 shows the relationship between the sound pressure level and the loudness level of human hearing.

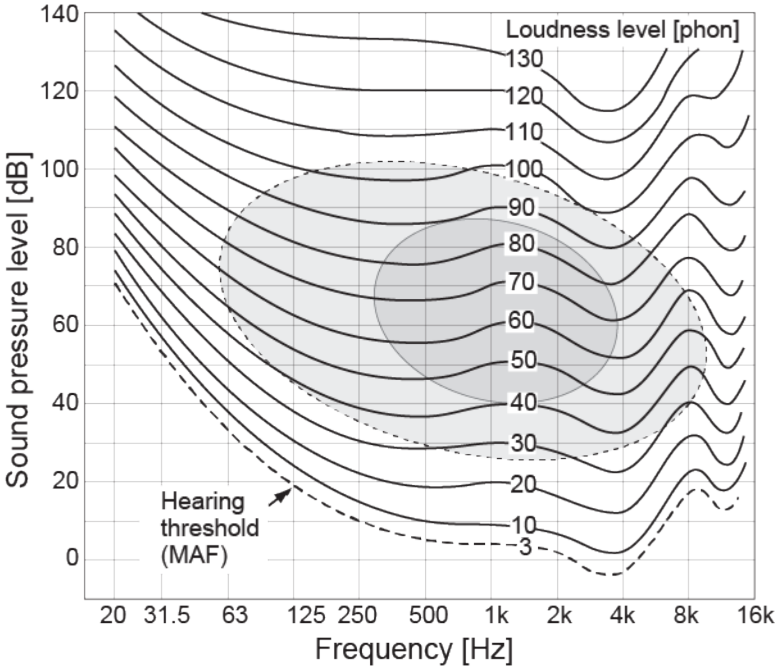

Figure 2.2

i) What is the rms pressure of a sound with SPL of 0 dB? Could a human hear this sound? (2 marks)

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ii) One wants to play pure tones of different frequencies <u>at the same loudness level</u>. If the sound pressure level of a 63 Hz tone is 90 dB, what sound pressure level does a 4 kHz tone have to be?

(1 mark)

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d) Three sets of head related impulse responses (HRIR) shown in Figure 2.3 were measured in an anechoic chamber by changing the position of a loudspeaker while a dummy head was fixed.

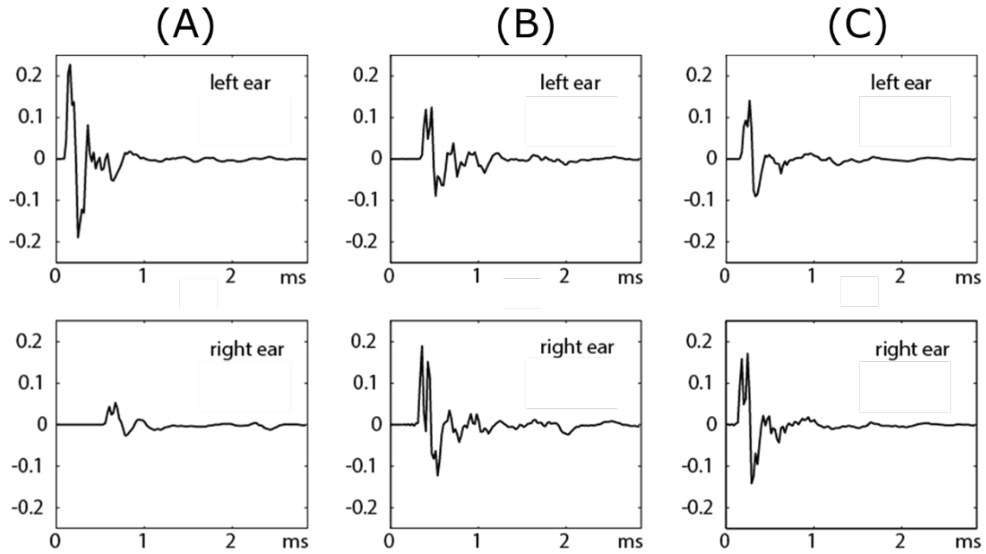
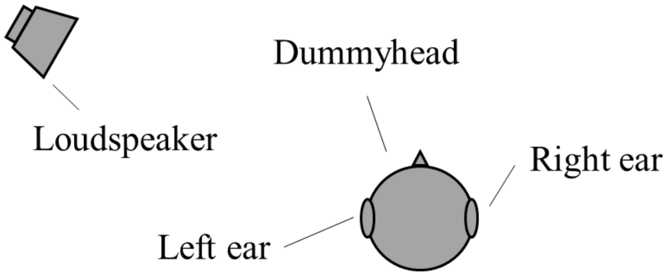


Figure 2.3

i) Choose the correct set of HRIR from (A) to (C) in Figure 2.3 corresponding to the case when the loudspeaker and dummy head were located as shown in the following diagram.

(1 mark)



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ii) The HRIR (B) and (C) were measured by placing the loudspeaker at different positions. When a sound is played from these two positions alternately, explain how the human hearing system will localise them correctly, <u>in no more than 100 words</u>.

(3 marks)

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Answer to Question 2.d.ii continued.

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3. a) i) What is required to produce sound?

(2 marks)

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ii) Give examples of two types of positive contribution that sound can make for humanity and also two types of negative contribution.

(2 marks)

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iii) When exposed to exactly the same sound field, one person describes it as a noise whilst another doesn't. Suggest two possible causes for the difference.

(1 mark)

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b) i) What functions of the peripheral ear contribute to raising the sensitivity of human hearing?

(3 marks)

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ii) The human cochlea is furnished with arrays of stereocilia or 'hair cells'. Explain how these function to provide us with the ability to analyse the frequency of sounds and extend the dynamic range of our hearing.

(2 marks)

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iii) A measure that has the symbol L_{AFmax} is sometimes used in noise limits. What phenomena of the human hearing system are reflected in this measure?

(2 marks)

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iv) Another measure, having the symbol L_{eq}, is also used in legislation. Explain what it is quantifying and why it is used in setting Damage Risk Criteria for human hearing protection.

(2 marks)

c) Workers at a fabrication factory have a working day which comprises four different activities. These activities don't vary significantly from day to day. Acoustics engineers sample these activities and find that representative samples of the sound levels and durations are as follows:

Duration (hrs)	L_{eq} (dB re 20 μPa)
2.00	90
1.50	80
1.50	85
2.00	83

For the remainder of their 8 hr shifts, the workers are taking refreshment breaks during which the sound levels are no higher than 60 dB.

i) Determine whether or not these workers are legally allowed to work without wearing hearing protection (e.g. ear muffs) if the legal limit is L_{eq} 85 dB for 8 hours.

(4 marks)

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Answer to question 3.c.i continued

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ii) By what percentage would their Noise Dose drop if the L_{eq} in the factory were reduced by 3 dB?

(2 marks)

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4. a) An acoustical engineer is required to assess the acoustic performance of two identical rooms which are separated by a common wall with a sound reduction index $R = 50$ dB. Both rooms are rectangular in shape and are unfurnished. The floor measures 10 m × 10 m and the walls are 3 m high.

i) The engineer measures the reverberation time in both rooms to be 0.5 s. Estimate the area-average sound absorption coefficient on the interior surfaces of each room.

(2 marks)

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ii) If the reverberant sound pressure level in one room (room 1) is 80 dB, calculate the reverberant SPL in the other room (room 2).

(3 marks)

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iii) The reverberant SPL calculated by the Acoustic Engineer in Q4.a.ii is too high. She therefore wishes to reduce the reverberant SPL in room 2 by 1 dB by attaching sheets of a sound absorbing material to the walls of room 2. This material has a sound absorption coefficient of 0.8. Calculate what area of this sound absorbing material should be added to room 2. Note that the walls of room 2 have a sound absorption coefficient of 0.1.

(5 marks)

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b) i) Calculate the sound pressure level measured by a microphone mounted on a rigid board on the ground 50 m away from an omni-directional source producing 1 Watt of acoustic power. You may assume that there are no excess attenuation effects other than the reflection on the rigid board.

(5 marks)

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ii) Estimate what the SPL would be at a microphone located 500 m from the source if this measurement is undertaken during a frost event and the microphone is mounted on a tripod 1.2 m above the ground.

(5 marks)

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5. A plane wave is incident from above on an infinite plane surface located in the $y = 0$ plane. The plane wave propagates at angle β to the x-axis. The complex pressure of the incident wave is denoted as p_i. The incident wave is reflected from the plane surface. The complex pressure associated with the reflected wave is denoted p_r. The total pressure in the air above the plane is the sum of the incident and reflected waves i.e.

$$p(x, y, t) = p_i(x, y, t) + p_r(x, y, t),$$

where

$$p_i(x, y, t) = A\exp\{\mathrm{i}\omega t - \mathrm{i}k\cos(\beta)\,x + \mathrm{i}k\sin(\beta)\,y\},$$

ω is the circular frequency of the wave, t is time, $k = \omega/c$ is the acoustic wavenumber, A is the amplitude of the incident wave and

$$p_r(x, y, t) = B\exp\{\mathrm{i}\omega t - \mathrm{i}k\cos(\beta)\,x - \mathrm{i}k\sin(\beta)\,y\},$$

where B is the amplitude of the reflected wave.

a) Show that the acoustic particle velocity in the y-direction on the $y = 0$ plane is given by

$$v = \sin(\beta)\frac{[B - A]}{\rho_0 c}\exp\{\mathrm{i}\omega t - \mathrm{i}k\cos(\beta)\,x\}.$$

(5 marks)

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Answer to question 5.a continued

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b) If the plane is rigid and $A = 1$ Pa, calculate the maximum and minimum SPL in the air above the plane. Hint: first derive an expression for the complex pressure, then an expression for the mean-square pressure, then calculate the maximum and minimum mean-square pressure and finally the maximum and minimum SPL.

(8 marks)

ID: ....

c) If the plane has impedance, z, show that the complex pressure above the plane is given by

$$p(x,y,t) = A\exp\{\mathrm{i}\omega t - \mathrm{i}k\cos(\beta)\,x\}\,[\exp\{\mathrm{i}k\sin(\beta)\,y\} + R\exp\{-\mathrm{i}k\sin(\beta)\,y\}],$$

where

$$R = \left(\frac{z\sin(\beta) + \rho_0 c}{z\sin(\beta) - \rho_0 c}\right).$$

(7 marks)

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Answer to question 5.c continued

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SEMESTER TWO 2020
Campus: City, Offshore Online

MECHANICAL ENGINEERING

Acoustics for Engineers

APPENDICES BOOKLET

Useful equations

Wave equation

$$\nabla^2 p - \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = 0$$

D'Alembert's solution.

$$p(x,t) = f^+(t - x/c) + f^-(t + x/c), \text{ (1D)}$$

$$p(x,t) = \frac{f^+\left(t - \frac{r}{c}\right)}{r} + \frac{f^-\left(t + \frac{r}{c}\right)}{r}, \text{ (Spherical wave)}$$

For a time-harmonic wave the acoustic pressure, $p(x,y,t)$, is related to the acoustic particle velocity in the x-direction, u, by the acoustic momentum equation:

$$\rho_0 \frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x}$$

For a time-harmonic wave the acoustic pressure, $p(x,y,t)$, is related to the acoustic particle velocity in the y-direction, v, by the acoustic momentum equation:

$$\rho_0 \frac{\partial v}{\partial t} = -\frac{\partial p}{\partial y}$$

Surface impedance is defined as the ratio of the complex pressure to the acoustic particle velocity normal to the surface at the surface i.e.

$$z = \frac{p}{u}$$

Mean-square pressure for time-harmonic waves with complex pressure p

$$\overline{p^2} = \frac{1}{2}pp^*$$

Relationships between frequency, f, wavelength, λ, and speed of sound, c.

$$c = f\lambda$$

Speed of sound in an ideal gas, $c = \sqrt{\gamma RT}$. For air the ratio of specific heats $\gamma = 1.4$, the gas constant $R = 287$ J/kg.K and zero Kelvins is equal to -273.15 °C.

Relationships between sound power, W, time-average radial acoustic intensity, I, and radius, r, for sound radiation from a point source.

$$I = W/4\pi r^2$$

Sound pressure level

$$\text{SPL} = 20\log_{10}\left(\frac{p_{\text{rms}}}{p_{\text{ref}}}\right), \qquad p_{\text{ref}} = 20 \times 10^{-6} \text{ Pa}$$

Sound intensity level

$$\text{IL} = 10\log_{10}\left(\frac{I}{I_{\text{ref}}}\right), \qquad I_{\text{ref}} = 10^{-12} \text{ W/m}^2$$

Sound power level

$$\text{SWL} = 10\log_{10}\left(\frac{W}{W_{\text{ref}}}\right), \qquad W_{\text{ref}} = 10^{-12}\ \text{W}$$

Combining levels for N incoherent sources

$$L = 10\log_{10}(10^{0.1L_1} + 10^{0.1L_2} + \cdots + 10^{0.1L_N})$$

Averaging levels for N incoherent sources

$$L = 10\log_{10}\left(\frac{10^{0.1L_1} + 10^{0.1L_2} + \cdots + 10^{0.1L_N}}{N}\right)$$

Time varying noise

$$L_{\text{eq,T}} = 10\log_{10}\left(\frac{T_1 10^{0.1L_1} + T_2 10^{0.1L_2} + \cdots + T_N 10^{0.1L_N}}{T_1 + T_2 + \cdots + T_N}\right)$$

Noise dose (for an 8 hour $L_{\text{A,eq,T}}$ of 85 dB)

$$\text{Dose} = 100\% \times 10^{(L_{\text{A,eq,T}} - 85)/10}$$

Octave band A-weighting values

Octave band centre frequency (Hz)	A-weighting value (dB)
63	-26
125	-16
250	-9
500	-3
1000	0
2000	1
4000	1
8000	-1

Sound absorption

$$T_{60} = 0.163\frac{V}{\alpha S}\ \text{(Sabine)}, \qquad T_{60} = -0.163\frac{V}{S\ln(1-\alpha)}\ \text{(Eyring)}$$

General insulation equation relating the sound pressure level (SPL) in room 1 to room 2 separated by a common wall of area S_w. Room 2 has total sound absorption $\alpha_2 S_2$.

$$\text{SPL}_2 = \text{SPL}_1 - R + 10\log_{10}(S_w) - 10\log_{10}(\alpha_2 S_2)$$

SPL measured a distance *r* away from a source with sound power level SWL with directivity index DI. The sound propagation is outdoors and is subject to excess attenuation A_E

$$\text{SPL} = \text{SWL} + \text{DI} - 10\log_{10}(4\pi r^2) - A_E$$